

Binary Ferrers rank-metric codes of distance three: Singleton attainment through area 21

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Abstract

We prove that every Ferrers diagram with at most 21 dots supports a binary linear matrix code attaining the three-cut Singleton bound for minimum nonzero rank at least three. The proof combines elementary constructions, a self-contained rectangular construction, fourteen explicit binary seed codes, and an exact finite reduction certificate. The certificate covers all 3,506 partitions of areas zero through 21, including the empty diagram. It uses 49 auxiliary diagrams and only inclusion, shortening, and transposition. All seed words and every reduction are checked with integer arithmetic. A separately implemented verification additionally reconstructs the codes throughout the reduction graph and checks all 1,389,655 codewords, counting the zero word once per code. The result concerns binary linear codes at distance three; the previously reported area-22 defect is cited as context and is not reverified here.

1 Statement and context

Use zero-based indices. For a finite weakly decreasing sequence $c = (c_0, \dots, c_{n-1})$ of positive integers, set

$$\mathcal{D}(c) = \{(r, j) : 0 \leq j < n, 0 \leq r < c_j\}, \quad |c| = \sum_{j=0}^{n-1} c_j.$$

We allow $c = ()$, with $\mathcal{D}(c) = \emptyset$. A supported matrix has zero entries outside $\mathcal{D}(c)$; for nonempty c its ambient size is $c_0 \times n$. Let $\kappa_2(c, 3)$ be the largest dimension of an $\mathbb{F}_2$ -linear subspace of these matrices in which every nonzero matrix has rank at least three. The zero code is allowed, with this condition holding vacuously. Define

$$\nu(c) = \min_{0 \leq i \leq 2} \nu_i(c), \quad \nu_i(c) = \sum_{j=2-i}^{n-1} \max(c_j - i, 0), \quad (1)$$

where an empty sum is zero. Thus ν_i counts the dots left after deleting the top i rows and the leftmost $2 - i$ columns. Horizontal reflection converts the usual top-right Ferrers convention, with rightmost-column deletion, to this top-left convention. It is an invertible column permutation and preserves rank, code dimension, and area. Empty padding rows and columns can be removed without changing these data.

Theorem 1. *For every finite weakly decreasing sequence c of positive integers with $|c| \leq 21$, including the empty sequence,*

$$\kappa_2(c, 3) = \nu(c).$$

Prajapati’s Theorem 1 in [2] gives a binary distance-three defect on $(5, 5, 5, 5, 1, 1)$: its area is 22, its bound is 12, and its largest code dimension is 11. Remark 8 there establishes attainment for the two smaller members E_4 and E_5 of the displayed family, under the heading “Minimality within the family.” The scope of Theorem 1 is attainment for *every* diagram of smaller area, rather than only that family. The area-22 nonexistence computation is not a dependency of our proof and has not been replayed here.

The elementary transport operations used below also appear in the irreducible-diagram framework of Couvée and Neri [1, Proposition 3.1]. We give all the transport and rectangular arguments needed here, so the proof does not depend on a specialized Ferrers existence theorem. The finite comparison in this paper is based on inspected primary material recorded on September 8–9, 2026; it is not a comprehensive priority assessment. As adjacent methodology, [3] concerns exact finite certification for a different matrix-support problem. That reference supplies no Ferrers-code existence result used here.

2 Bounds, constructions, and transport

Lemma 2 (Three-cut bound). *For every c , $\kappa_2(c, 3) \leq \nu(c)$.*

Proof. Fix $i \in \{0, 1, 2\}$ and restrict matrices to the positions counted by $\nu_i(c)$. A matrix in the kernel is supported on the union of i rows and $2 - i$ columns. Split it into its entries on those rows and its remaining entries on those columns. The two summands have rank at most i and $2 - i$, respectively. The kernel therefore contains no nonzero word of a code whose minimum nonzero rank is at least three. Restriction is injective on the code, giving dimension at most $\nu_i(c)$. Minimize over i . $\square$

Call a diagram *attaining* if it supports such a code of dimension $\nu(c)$.

Lemma 3 (Transport of attainment). *Each of the following operations transports attainment from a to b :*

1. $\mathcal{D}(a) \subset \mathcal{D}(b)$, $|b| = |a| + 1$, and $\nu(b) = \nu(a)$;
2. $\mathcal{D}(b) \subset \mathcal{D}(a)$, $|b| = |a| - 1$, and $\nu(b) = \nu(a) - 1$;
3. $b = a^\top$, the conjugate partition.

Proof. For the first operation, use the same code with zero padding. For the second, take the kernel of the deleted coordinate on an attaining code on a . Its dimension is at least $\nu(a) - 1 = \nu(b)$, its support is in $\mathcal{D}(b)$, and all its nonzero ranks are still at least three. Lemma 2 forces its dimension to equal $\nu(b)$. For the third, transpose every matrix. This preserves rank and dimension and exchanges cut i with cut $2 - i$ in (1). Exterior all-zero rows or columns may be discarded in either of the first two operations. $\square$

Lemma 4 (Elementary and rectangular roots). *Every diagram with $\nu(c) \leq 1$ is attaining, as is every rectangle.*

Proof. For $\nu(c) = 0$ use the zero code. If $\nu(c) = 1$, positivity of all three cuts forces at least three columns, $c_1 \geq 2$, and $c_0 \geq 3$. Consequently $(2, 0)$, $(1, 1)$, $(0, 2)$ are present. Ones in these three positions and zeros elsewhere give a rank-three matrix whose span attains the bound.

For an $m \times n$ rectangle put $a = \max(m, n)$ and $b = \min(m, n)$. If $b < 3$, the bound is zero. Otherwise take $K = \mathbb{F}_{2^a}$ and b elements $x_1, \dots, x_b$ independent over $\mathbb{F}_2$. For coefficients $A_0, \dots, A_{b-3} \in K$ consider

$$f(X) = \sum_{t=0}^{b-3} A_t X^{2^t}. \quad (2)$$

Expand $(f(x_1), \dots, f(x_b))$ in an $\mathbb{F}_2$ -basis of K to obtain an $a \times b$ binary matrix. A nonzero coefficient tuple gives a nonzero ordinary polynomial of degree at most 2^{b-3} . It has at most that many roots in K . Since f is $\mathbb{F}_2$ -linear, its kernel on the b -dimensional span of the x_j has dimension at most $b - 3$. The matrix has rank at least three. This also proves that the coefficient-to-matrix map is injective, so the code dimension is $a(b - 2)$. The other two rectangular cuts exceed $a(b - 2)$ by $a - b + 1$ and $2(a - b)$, respectively. Thus this is the minimum cut. Transpose the matrices if necessary to obtain the desired orientation. $\square$

3 Exact seeds and complete finite coverage

The remaining roots are the fourteen codes in Table 1. Appendix A prints their full generator lists, also supplied in machine-readable form. In each list an integer v_j encodes column j : its bit r is the entry in row r , with bit zero at the top. A row of the printed list specifies one entire generator matrix, rather than one row of that matrix.

Seed	Column heights	ν	Words checked
1	(3, 3, 1, 1)	2	4
2	(4, 3, 3, 1)	4	16
3	(4, 4, 4, 1, 1)	6	64
4	(5, 4, 4, 4, 1)	9	512
5	(5, 5, 5, 4, 1, 1)	11	2048
6	(6, 5, 3, 3, 3, 1)	10	1024
7	(6, 5, 4, 3, 2, 1)	10	1024
8	(6, 6, 3, 2, 2, 1, 1)	9	512
9	(6, 6, 3, 2, 2, 2)	9	512
10	(6, 6, 3, 3, 1, 1, 1)	9	512
11	(6, 6, 4, 2, 1, 1, 1)	9	512
12	(6, 6, 5, 1, 1, 1, 1)	9	512
13	(6, 6, 6, 1, 1, 1)	9	512
14	(7, 7, 2, 1, 1, 1, 1, 1)	7	128

Table 1: The fourteen explicit roots. Word counts include the zero word.

For each list of k generators, enumerate all 2^k binary coefficient vectors and form the resulting column integers by bitwise exclusive-or. For each word check $0 \leq v_j < 2^{c_j}$. To compute rank, start with $S = \{0\}$ and, for each column v_j , replace S by $S \cup \{s \oplus v_j : s \in S\}$. This is precisely the binary column span, so the rank is $\log_2 |S|$. Check that every nonzero coefficient vector gives rank at least three, that the 2^k words are distinct, and that $k = \nu(c)$. These finite checks succeed for all fourteen lists: 7,892 words are checked in total. The supplement contains the checker and exact data, so the calculation requires no discovery search. In particular the displayed basis for (3, 3, 1, 1) is the basis checked here; we do not assert that it reproduces a displayed basis in [2] verbatim.

Proposition 5 (Finite certificate). *The supplied reduction certificate covers every partition of area at most 21. Each route starts at a root of Lemma 4 or at one of the fourteen verified codes, and uses only the forward operations of Lemma 3.*

Exact finite verification. The certificate is a list of 3,555 distinct partitions, with their three cuts and either a root type or a parent and operation type. The checker recomputes cuts by counting dots, rather than trusting the stored values. It verifies every one-dot containment and its prescribed change of ν , and every transpose relation. A depth-first traversal with an active-vertex set rejects a cycle or a missing parent. Every path terminates at a verified root.

The longest path has fifteen edges. The record counts are

elementary roots	675	inclusions	2431
rectangular roots	21	shortenings	309
explicit roots	14	transpositions	105.

These counts sum to 3,555. Exactly 49 vertices have area greater than 21; all vertices have area at most 30.

Independently of the route list, the checker enumerates partitions as follows. For remaining area N and smallest allowed part s , choose $p \in \{s, \dots, N\}$, recursively partition $N - p$ with all parts at least p , and append p to the decreasing list returned by the recursion. The base case $N = 0$ returns the empty partition. Every nonempty partition has a unique smallest part and unique remainder, so induction proves that this enumerates every partition exactly once. The checker applies this recursion for $N = 0, \dots, 21$ and verifies that each result occurs in the route list. The numbers at the successive areas are

$$1, 1, 2, 3, 5, 7, 11, 15, 22, 30, 42, \\ 56, 77, 101, 135, 176, 231, 297, 385, 490, 627, 792,$$

with sum 3,506. As a redundant check these counts agree with the coefficients through degree 21 of $\prod_{j=1}^{21} (1 - x^j)^{-1}$. All route, root, and coverage checks succeed on the supplied files. $\square$

Proof of Theorem 1. By Proposition 5, every target partition has a finite route from an attaining root. Induction along that route using Lemma 3 constructs an attaining code. Together with Lemma 2, this proves the stated equality. $\square$

4 Reproduction and scope of the computation

The accompanying **supplement** directory contains the exact seed and route data and a standard-library Python verifier. From that directory, run the following command using ordinary Python 3 (without optimization):

```
python3 evidence/verify_final.py
```

The inputs are **source.md** and the following two files:

```
evidence/final-seed-certificates.json
evidence/final-coverage.json
```

The output is **evidence/final-verification.json**. The source text is included to preserve the original question; its title for the motivating paper is a descriptive variant, while the bibliography below uses the primary title. The JSON data are explicit finite proof certificates. Both checking the seeds and checking coverage are essential: numerical counts alone would not establish Proposition 5.

A separately written verifier, importing none of the discovery or original verification code, also rebuilt the partition list by adding Ferrers corners, checked the counts by Euler’s pentagonal recurrence, and used binary row elimination for ranks. It reconstructed codes at all 3,555 vertices, including rectangular roots over explicitly checked binary extension fields and coordinate kernels for shortenings. Support, generator independence, dimension, and every word rank passed. There were 1,389,655 words across all graph codes, of which 248,055 belong to the 3,506 target codes. Each total includes one zero word per code; the totals do not count distinct matrices across different ambient spaces. This additional replay corroborates the explicit finite argument above.

The target is exactly binary linear codes with minimum nonzero rank at least three and area at most 21. The auxiliary area-30 cap is a property of this certificate, not a claim of attainment for every diagram through area 30. We make no assertion for nonlinear codes, other fields or distances, or independent verification of the cited area-22 upper bound. Individual seed codes are not each claimed as separate contributions.

Computational and AI assistance. AI assistance was used to develop the arguments, generate and run exact search and checking programs, and prepare this manuscript. Randomness was used in one discovery search order, with fixed seed 20260909; all submitted certificates and verification steps are deterministic. A separate AI model context performed the mathematical review and additional implementation described above. This is an implementation and review separation, not external peer review or a claim of statistical independence. No floating-point test, solver timeout, or unsatisfiability claim is used to prove Theorem 1.

References

- [1] H. Beeloo-Sauerbier Couvée and A. Neri, *Irreducible Ferrers diagrams in the Etzion–Silberstein conjecture*, arXiv:2604.27868v1 (2026). <https://arxiv.org/abs/2604.27868v1>.
- [2] J. Prajapati, *A counterexample to the Etzion–Silberstein conjecture*, arXiv:2608.08478v2 (2026). <https://arxiv.org/abs/2608.08478v2>.
- [3] Z. Zeng, *Exact Finite Certification of The Sparse-Complement Two-Eigenvalue Conjecture at Order Nine*, preprint (2026). <https://doi.org/10.2139/ssrn.7379578>.

A Complete binary generator lists

The encoding and numbering are those of Table 1. Each row below is a tuple of column integers encoding one generator matrix.

Seed 1: (3, 3, 1, 1)

2	4	1	0
4	6	0	1

Seed 2: (4, 3, 3, 1)

4	2	1	0
8	4	0	1
3	4	2	0
6	3	4	0

Seed 3: (4, 4, 4, 1, 1)

2	4	1	0	0
4	8	0	1	0
8	6	0	0	1
12	1	2	0	0
11	9	4	0	0
1	2	8	0	0

Seed 4: (5, 4, 4, 4, 1)

5	9	1	0	0
23	10	0	1	0
10	15	0	0	1
19	12	2	0	0
15	3	0	2	0
27	7	4	0	0
22	13	0	4	0
15	4	8	0	0
27	5	0	8	0

Seed 5: (5, 5, 5, 4, 1, 1)

2	4	1	0	0	0
4	8	0	1	0	0
8	16	0	0	1	0
16	6	0	0	0	1
5	1	2	0	0	0
12	4	0	2	0	0
28	10	4	0	0	0
2	7	0	4	0	0
4	19	8	0	0	0
1	6	0	8	0	0
13	7	16	0	0	0

Seed 6: (6, 5, 3, 3, 3, 1)

2	4	1	0	0	0
4	8	0	1	0	0
8	16	0	0	1	0
16	6	0	0	0	1
5	1	2	0	0	0
12	4	0	2	0	0
32	8	0	0	2	0
8	3	4	0	0	0
48	1	0	4	0	0
41	17	0	0	4	0

Seed 7: (6, 5, 4, 3, 2, 1)

2	4	1	0	0	0
4	8	0	1	0	0
8	16	0	0	1	0
16	6	0	0	0	1
5	1	2	0	0	0
12	4	0	2	0	0
32	8	0	0	2	0
8	3	4	0	0	0
18	1	0	4	0	0
14	16	8	0	0	0

Seed 8: (6, 6, 3, 2, 2, 1, 1)

2	4	1	0	0	0	0
4	8	0	1	0	0	0
8	16	0	0	1	0	0
16	32	0	0	0	1	0
32	6	0	0	0	0	1
3	8	2	0	0	0	0
8	5	0	2	0	0	0
12	32	0	0	2	0	0
8	18	4	0	0	0	0

Seed 9: (6, 6, 3, 2, 2, 2)

2	4	1	0	0	0
4	8	0	1	0	0
8	16	0	0	1	0
16	6	0	0	0	1
5	1	2	0	0	0
12	4	0	2	0	0
32	8	0	0	2	0
13	32	0	0	0	2
8	3	4	0	0	0

Seed 10: (6, 6, 3, 3, 1, 1, 1)

2	4	1	0	0	0	0
4	8	0	1	0	0	0
8	16	0	0	1	0	0
16	32	0	0	0	1	0
32	6	0	0	0	0	1
3	8	2	0	0	0	0
8	5	0	2	0	0	0
8	18	4	0	0	0	0
6	32	0	4	0	0	0

Seed 11: (6, 6, 4, 2, 1, 1, 1)

2	4	1	0	0	0	0
4	8	0	1	0	0	0
8	16	0	0	1	0	0
16	32	0	0	0	1	0
32	6	0	0	0	0	1
3	8	2	0	0	0	0
8	5	0	2	0	0	0
8	18	4	0	0	0	0
12	28	8	0	0	0	0

Seed 12: (6, 6, 5, 1, 1, 1, 1)

2	4	1	0	0	0	0
4	8	0	1	0	0	0
8	16	0	0	1	0	0
16	32	0	0	0	1	0
32	6	0	0	0	0	1
3	8	2	0	0	0	0
8	6	4	0	0	0	0
12	18	8	0	0	0	0
32	3	16	0	0	0	0

Seed 13: (6, 6, 6, 1, 1, 1)

2	4	1	0	0	0
4	8	0	1	0	0
8	16	0	0	1	0
16	6	0	0	0	1
5	1	2	0	0	0
8	3	4	0	0	0
32	16	8	0	0	0
3	32	16	0	0	0
25	24	32	0	0	0

Seed 14: (7, 7, 2, 1, 1, 1, 1)

2	4	1	0	0	0	0
4	8	0	1	0	0	0
8	16	0	0	1	0	0
16	32	0	0	0	1	0
32	64	0	0	0	0	1
64	6	0	0	0	0	1
3	8	2	0	0	0	0